THE CHINESE UNIVERSITY OF HONG KONG
Department of Statistics and Data Science

STAT3003: Survey Methods
Problem Sheet 4

The deadline for submission is **9:30am** on 4 December 2025 (Thursday). Please submit your solutions in soft copy via the link provided on the course Blackboard page. **No late submissions will be accepted. A late submission will receive a mark of zero.** Students may discuss set problems with others, but their final submissions must be their own work.

Please answer the following problems. Show your working.

1. (Adapted from Exercise 2.21 of Scheaffer et al. (2012)) Readers of the magazine *Popular Science* (August 1990) were asked to phone in their responses to the following question: "Should the U.S. build more fossil-fuel generating plants or the new so-called safe nuclear generators to meet the energy crisis of the 90s?". Of the total call-ins, 86% chose the nuclear option. What do you think about the way the poll was conducted? What do you think about the way the question was worded? Do you the results are a good estimate of the prevailing mood of the country?

2. (Adapted from Exercise 2.7 of Scheaffer et al. (2012)) Discuss the relative merits of using personal interviews, telephone interviews, and mailed questionnaires as methods of data collection for each of the following situations:

   (a) A television executive wants to estimate the proportion of viewers in the country who are watching her network at a certain hour.

   (b) A newspaper editor wants to survey the attitudes of the public toward the type of news coverage offered by his paper.

   (c) A district councillor is interested in determining how homeowners feel about a proposed zoning change (from "industrial" to "residential", say).

   (d) The Department of Health wants to estimate the proportion of dogs that have had vaccinations for rabies within the last year.

3. (Adapted from Exercise 2.10 of Scheaffer et al. (2012)) Give an example of a question that could force a response in a certain direction because of its strong wording. *Note: please do not use an example already given in the notes or exercises or previous solutions...*

4. (Adapted from Exercise 2.31 of Scheaffer et al. (2012)) Balance of questions makes logical sense but does not always have a strong impact on the results. Its impact is increased by the strength of a counterargument. The following two comparisons were drawn in one study reported by Schuman and Presser (1996):

> A1: If there is a serious fuel shortage this winter, do you think there should be a law requiring people to lower the heat in their homes?
>
> B1: If there is a serious fuel shortage this winter, do you think there should be a law requiring people to lower the heat in their homes, or do you oppose such a law?

Another study in a later year compared these forms:

> A2: If there is a serious fuel shortage this winter, do you think there should be a law requiring people to lower the heat in their homes, or do you oppose such a law? (Same as B1 form).
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> B2: If there is a serious fuel shortage this winter, do you think there should be a law requiring people to lower the heat in their homes, or do you oppose such a law because it would be too difficult to enforce?

Within each pair of questions, which one do you think received the lower percentage of responses favouring the law? Explain your reasoning.

5. (Adapted from Exercise 11.10 from Scheaffer et al. (2012)) A corporation executive wants to estimate the proportion of corporation employees who have been convicted of a misdemeanor. Because the employees would not want to answer the question directly, the executive uses a random-response technique. A SRS of 300 people is selected from a large number of corporation employees. In separate interviews, each employee draw a card from a deck that has 0.7 of the card marked "convicted" and 0.3 marked "not convicted." The employee obtains 115 "yes" responses. Estimated the proportion of employees who have been convicted of a misdemeanor and provide a 95% confidence interval.

6. (Adapted from Exercise 5.21 from Scheaffer et al. (2012)) A quality control inspector must estimate the proportion of defective microcomputer chips coming from two different assembly operations. She knows that, among the chips in the lot to be inspected, 60% are from assembly operation A and 40% are from assembly operation B. In a SRS of 100 chips, 36 turn out be from operation A and 64 from operation B. Among the sampled chips from operation A, six are defective. Among the sampled chips from operation B, ten are defective.

   (a) Considering only the SRS of 100 chips, estimate the proportion of defectives in the lot, and provide a 95% confidence interval.

   (b) Stratifying the sample, after selection, into chips from operation A and B, estimate the proportion of defectives in the population, and estimate the standard error. Ignore the finite population correction in both cases. Which answers do you find more acceptable?

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